

VARIABLE AREA FLOWMETER

가변 면적 유량계

KVA-500 Series



VARIABLE AREA FLOWMETER

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KVA-500
S E R I E S

K-METER

액체 및 가스 유량 측정용



VARIABLE AREA
FLOWMETER

특징 FEATURE

- 전체 금속 구조로 제작되어 내구성이 뛰어남
- 전도도, 유전율 등 매체 상태에 영향을 받지 않음
- 밀도가 다른 두 매체의 계면 또는 레벨 측정 가능
- 부식성, 독성 등 다양한 매체 적용 가능
- ANSI, DIN, KS 등 다양한 Connection 제공
- 4~20mA 출력 및 HART, 경보 출력 옵션



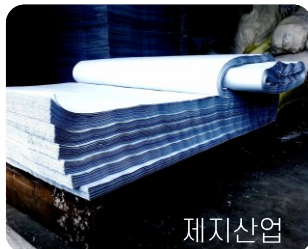
수처리시설



제약산업



식품 공정



제지산업



화학 및 기타 공업



관개

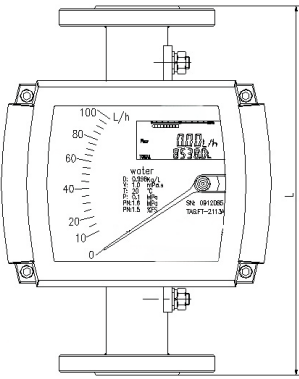
적용 산업 APPLICATIONS

- 하수, 오수 상하수도 수처리시설
- 석유화학 및 기타 공업
- 화학물질 모니터링
- 식품 공정
- 제지산업
- 제약산업
- 야금산업
- 관개시설

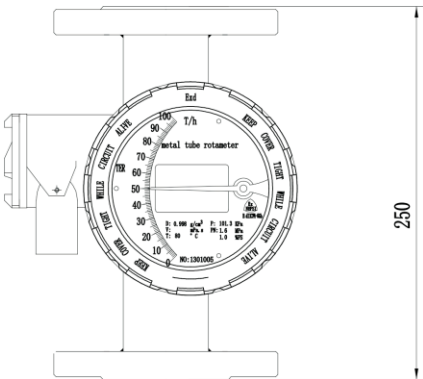
가변 면적 유량계 KVA-500 시리즈	
구 분	사 양 정 보
지시부 타입 (Indicator Type)	포인터형 (Local-Pointer Type) / 디지털 디스플레이형 (Digital LCD Type)
유량 측정 범위 (Measuring Range)	Water (20℃) : 16 ~ 150000 l/h Gas (0.1013 Mpa, 20℃) : 0.5 ~ 4000 m3/h
측정 가능 배관 사이즈 (Measuring Pipe Size)	15 mm ~ 150 mm Process Connection : Flange (DIN, ANSI, JIS), Tri-clamp, Thread (BSP, NPT)
측정 가능 범위 비율 (Turn-down Ratio)	10:1 (Special type 20 : 1)
정확도 등급 (Accuracy Class)	2.5 (Special type 1.5% or 1.0%)
작동 압력 (Working Pressure)	DN15 ~ DN50 1.6 MPa DN80 ~ DN150 1.0 MPa (Special type 1.6 MPa) Pressure rating of jacket 1.6MPa
온도 (Temperature)	Standard: -80℃ ~ +220℃ (Digital display with 4~20 mA) High temperature: 300℃ (Mechanical with Indicator) FEP liner type 90℃
주변 온도 (Ambient Temperature)	-20℃~+100℃ (remote type without LCD display ≤85℃) (remote type with LCD display ≤70℃)
점도 (Viscosity)	DN15: ≤30 mPa.s DN25: ≤250 mPa.s DN50 ~ DN150 : ≤300 mPa.s
출력 (Output)	Standard : Two-wire 4~20 mA (HART optional), Three-wire 0~10 mA Pulse
통신 (Communication)	RS-485(Modbus RTU), HART
알람 신호 (Alarm Signal)	Two relay outputs (Limits 125VAC/0.25A) One or two proximity switches Pulse output : 0 ~ 1KHz, Isolated output (Output Level Vpp > 4.5V)
전원 (Power Supply)	Standard: 24 VDC±20% Customized: 220 VAC (85~265 VAC) Battery powered: 3.7@4.4~5.2 AH Lithium Battery, 12~36 months.
보호등급 (Protection Class)	IP65 / IP67
센서부 (Wetted Parts)	SS304 SS316L, FEP, Hastelloy C, Titanium

외관 치수 DIMENSION

◎ Standard Type : Dimensions and Weight



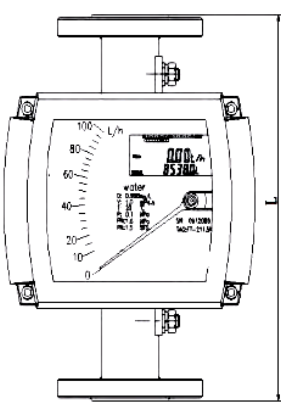
Square Converter



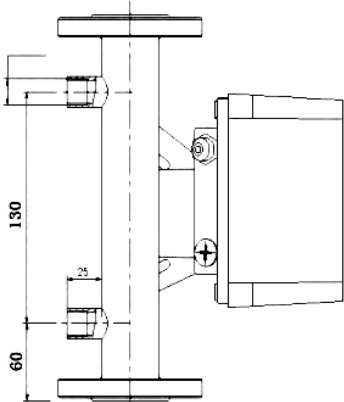
Round Converter

Caliber	L (mm)	Weight (kg)
DN15	250	5.0
DN25	250	6.5
DN50	250	10
DN80	250	15.5
DN100	250	17
DN150	250	35

◎ Jacket Type : Dimensions and Weight

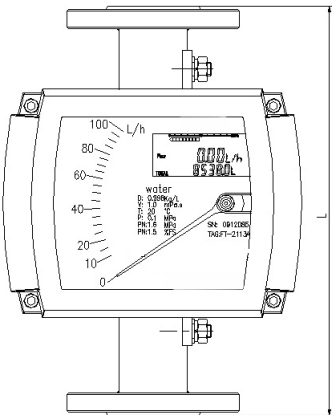


Insulation Jacket type

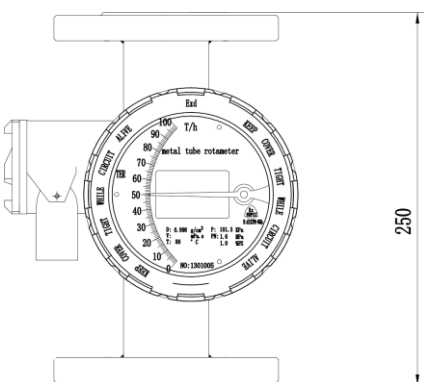


Caliber	L (mm)	Weight (kg)
DN15	250	7.5
DN25	250	10
DN50	250	13
DN80	250	19
DN100	250	21
DN150	250	38

◎ FEP Liner Type : Dimensions and Weight



Square Converter

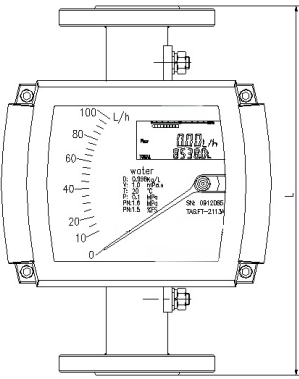


Round Converter

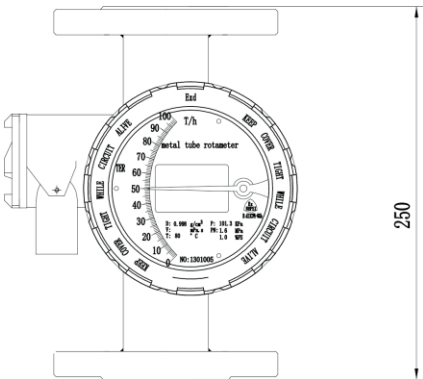
Caliber	L (mm)	Weight (kg)
DN15	250	5.0
DN25	250	6.5
DN50	250	10
DN80	250	15.5
DN100	250	16.5
DN150	250	32

외관 치수 DIMENSION

◎ Vertical Outlet Type : Dimensions and Weight



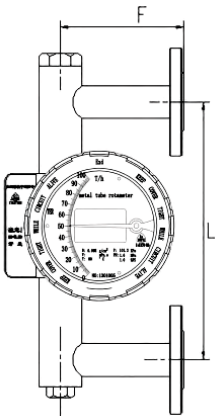
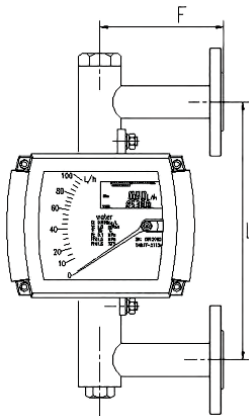
Square Convertor



Round Convertor

Caliber	L (mm)	Weight (kg)
DN15	250	5.0
DN25	250	6.5
DN50	250	10
DN80	250	15.5
DN100	250	17
DN150	250	35

◎ Side Outlet Type : Dimensions, Weight and Pressure Loss

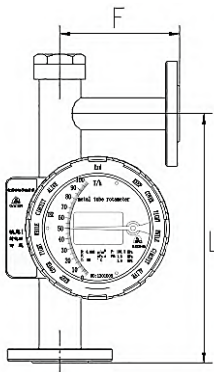
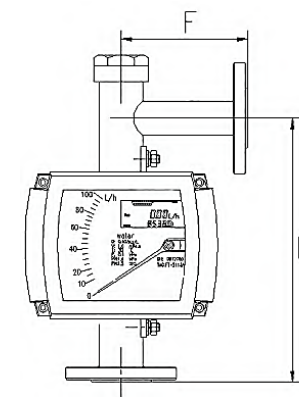


Caliber	DN15	DN25
F(mm)	120	120
L(mm)	250	250
Weight(kg)	6	7.2
Pressure loss(kpa)	21	30

(DN15 ~ DN25)

DN32 ~ DN150 drawing can be provided on request

◎ Bottom Inlet and Side Outlet Type: Dimensions, Weight and Pressure Loss



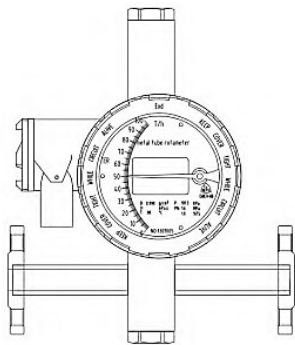
Caliber	DN15	DN25
F(mm)	120	120
L(mm)	250	250
H(mm)	350	350
Weight(kg)	4.5	7
Pressure loss(kpa)	18	22

(DN15 ~ DN25)

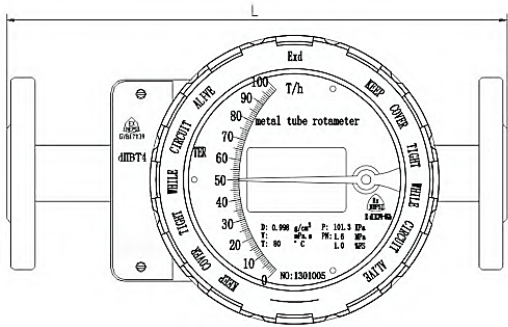
DN32 ~ DN150 drawing can be provided on request

외관 치수 DIMENSION

◎ Horizontal Mounting Type : Dimensions, Weight and Pressure Loss



Gas DN15 ~ DN150

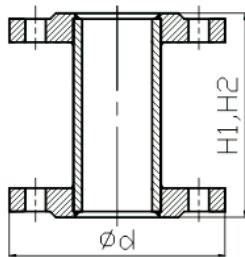


Liquid DN15 ~ DN15

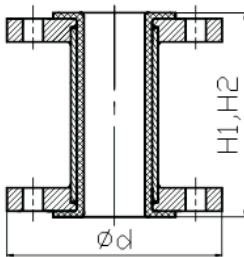
Caliber	L (mm)
DN15	250
DN20	250
DN25	250
DN40	300
DN50	300
DN65	400
DN80	400
DN100	400
DN125	500
DN150	500

Caliber	L (mm)
DN15	250
DN20	250
DN25	250
DN40	250
DN50	250
DN65	250
DN80	250
DN100	250
DN125	250
DN150	250

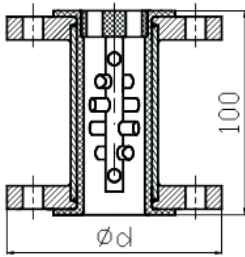
◎ Additional Structure and Installation Instructions



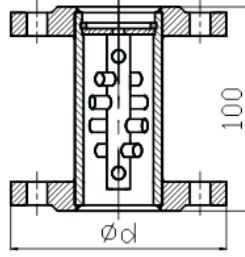
Straight pipe type



Liner FEP straight pipe type



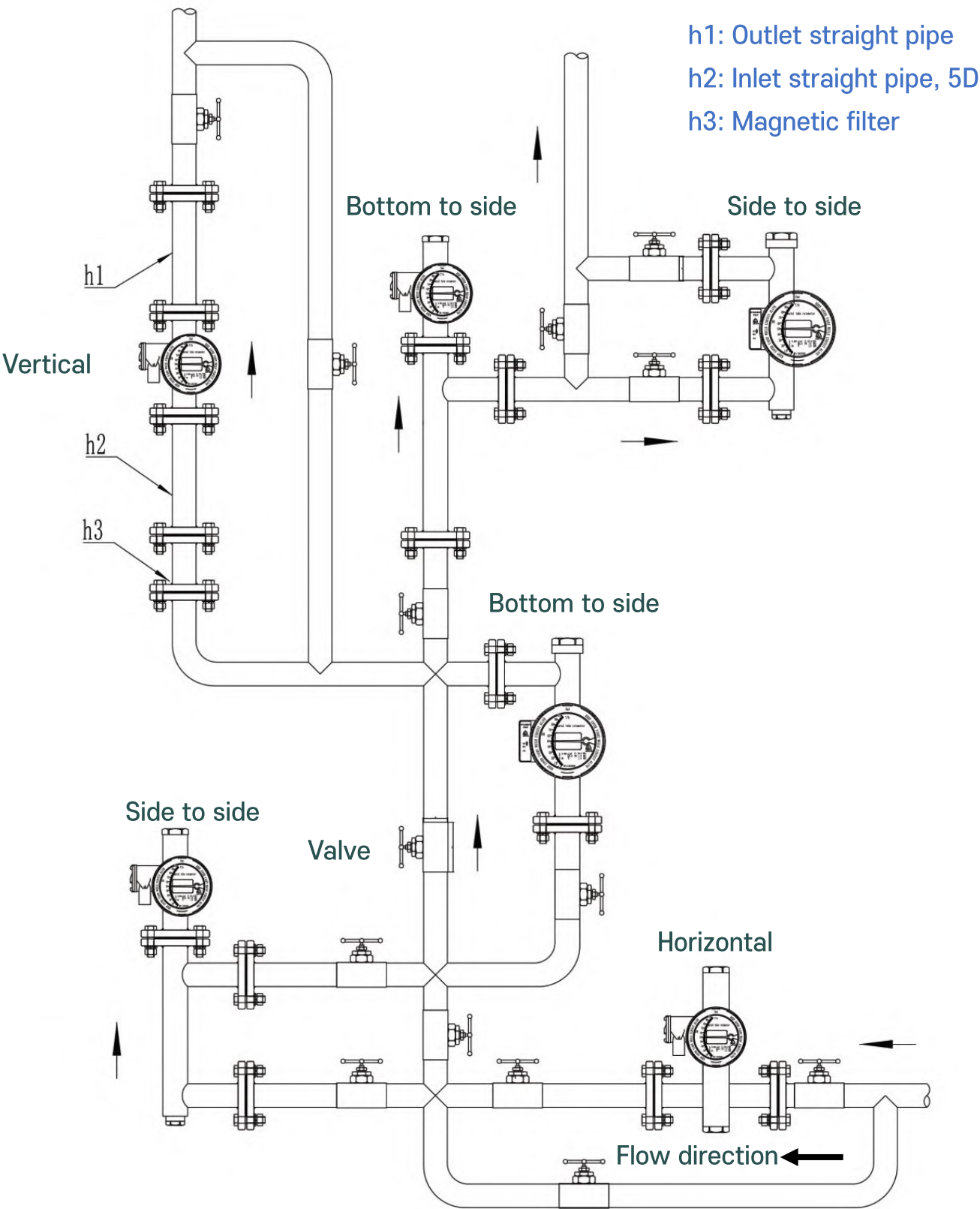
Liner FEP with Magnetic Filter



With Magnetic Filter

Diameter	DN15	DN25	DN50	DN80	DN100	DN150
Front straight pipe H1≥(mm)	75	125	250	400	120	750
After straight pipe H2≥(mm)	250	250	250	250	250	250
φd (mm)	95	115	165	200	350	285

설치 방법 Intallation Drawing



유량표 FLOW RANGE TABLE

VARIABLE AREA FLOWMETER KVA-500 Series

Diameter DN	Water (20 ℃) L/h		Air 0.1013MPa 20℃ Nm3/h	Standard type Max pressure loss Kpa	
	Standard	FEP	Standard	Water	Air
15	16	-	0.5	2.0	7.0
	25	16	0.7	2.3	7.2
	40	25	1.1	2.5	7.3
	63	40	1.8	2.5	7.5
	100	63	2.8	2.5	7.8
	160	100	4.8	2.6	8.0
	250	160	7.0	2.7	10.0
	400	250	10.0	2.9	10.8
	600	400	16.0	3.4	14
20	600	400	16	4.0	7.0
	1000	600	30	4.1	8.0
	1600	1000	45	4.4	12.0
	2500	1600	70	5.2	19.0
	4000	2500	110	7.0	25.0
	6000	4000	180	12.5	33.0
25	600	400	16	4.0	7.0
	1000	600	30	4.1	8.0
	1600	1000	45	4.4	12.0
	2500	1600	70	5.2	19.0
	4000	2500	110	7.0	25.0
	6000	4000	180	12.5	33.0
32	1000	600	30	4.1	8.0
	1600	1000	4.5	4.4	12.0
	2500	1600	70	5.2	19.0
	4000	2500	110	7.0	25.0
	6000	4000	180	12.5	33.0
	10000	-	250	12.5	33.0

유량표 FLOW RANGE TABLE

VARIABLE AREA FLOWMETER KVA-500 Series

Diameter DN	Water (20 ℃) L/h		Air 0.1013MPa 20℃ Nm3/h	Standard type Max pressure loss Kpa	
	Standard	FEP	Standard	Water	Air
40	2500	1600	70	5.2	19.0
	4000	2500	110	7.0	25.0
	6000	4000	180	12.5	33.0
	10000	-	250	12.5	33.0
50	6000	4000	180	4.7	8.0
	10000	6000	250	5.1	15.0
	16000	10000	400	6.2	22.0
	25000	16000	600	8.0	35.0
65	16000	10000	400	6.2	22.0
	25000	16000	600	8.0	35.0
80	25000	25000	1000	5.3	15.0
	40000	40000	1200	7.8	22.0
	60000	40000	1600	8.3	25.0
100	60000	40000	1800	11.4	35.0
	100000	60000	3000	16.7	45.0
125	100000	40000	3000	11.4	42.0
	125000	50000	3000	11.4	47.0
150	150000	100000	4000	17.0	47.0

비고: 본 표에 기재된 유량은 표준 유량입니다.
고객 요청에 따라 별도 사양으로 맞춤 제작이 가능합니다.

모델 코드표 MODEL SELECTION GUIDE

VARIABLE AREA FLOWMETER

KVA-500 Series														
KVA - 500 -		x	x	x	x	x	x	x	x	x	x	x	x	x
Size	DN15 ~ DN150 (1/2" ~ 6")													
Indicator	Mechanical	M												
	Digital LCD display	D												
Installation	Vertical	V												
	Horizontal	H												
Structure	Bottom-top	BT												
	Top-bottom	TB												
	Left-right (horizontal)	LR												
	Right-left (horizontal)	RL												
	Side-side	SS												
	Bottom-side	BS												
Accuracy	±2.5%	A25												
	±1.5%	A15												
Temperature	-20°C ~ 200°C	T2												
	-20°C ~ 300°C	T3												
Wetted parts	SS304	S4												
	SS304 lined FEP	S4F												
	SS316L	S6												
	Hastelloy C	HC												
	Titanium	Ti												
Power supply	No	SN												
	24 VDC	DC												
	220 VAC	AC												
	3.6 V lithium battery	BA												
Signal output	No	SN												
	4~20 mA	C												
	RS485	R												
	HART	H												
	Pulse	P												
Process connection	Flange	D10: DIN PN10, D16: DIN PN16, D25: DIN PN25, D40: DIN PN40										DXX		
		A15: ANSI 150#, A30: ANSI 300#										JXX		
		J10: JIS 10K, J20: JIS 20K, J30: JIS 30K										AXX		
	Tri-clamp											TC		
	Thread											T		
Alarm	No												AN	
	High												H	
	Low												L	
	High + Low												HL	
Ex-proof	No													EN
	Flame-proof (Ex d IIC T4~T6 Gb)													Exd
	Intrinsic safe (Ex ia IIC T3~T6)													Exia
Additional structure	No													ASN
	Damp													AD
	Jacket													AJ

제품 및 검교정 문의 연락처

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E-MAIL : info@flowmeters.co.kr

본사 : 인천광역시 부평구 안남로 425-10



(주)엠앤씨 엔지니어링
M&C Engineering Inc.



2025_KVA500_R02

정확한 계측과 신뢰도 높은 측정 장치의 기준, 엠앤씨 엔지니어링이 앞장서겠습니다.

www.flowmeters.co.kr / www.kmeter.co.kr



INNOBIZ

KMETER

KASTO